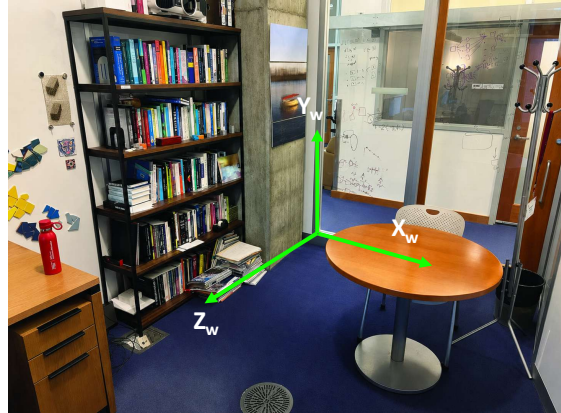


Agenda

- Dissecting the camera matrix



1

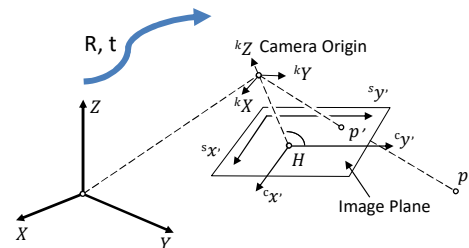
Recap

- Computer Vision camera is a matrix: What?

$$x_{3 \times 1} = k_{3 \times 3} [I_{3 \times 3} | 0] X_{cam} = K_{3 \times 3} R [I_{3 \times 3} | -\hat{C}] X_w$$

$$x_{3 \times 1} = K_{3 \times 3} R [I_{3 \times 3} | -\hat{C}] X_w = PX$$

$$P = KR[I | -\hat{C}] = K[Rt]$$

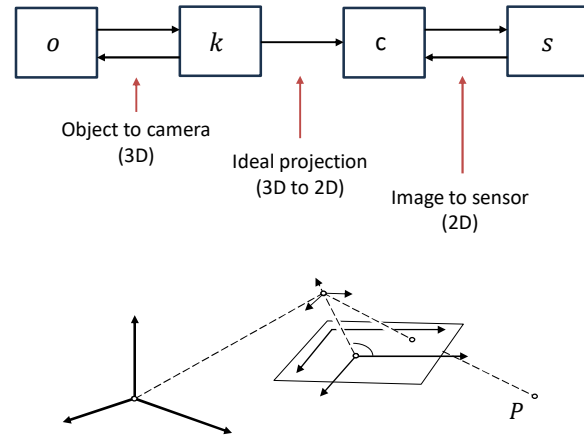


$$K = \begin{bmatrix} \alpha_x & s & m_x p_x \\ & \alpha_y & m_y p_y \\ & & 1 \end{bmatrix}$$

2

World to Sensor

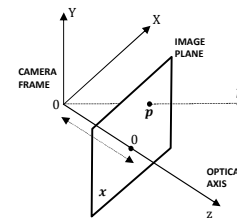
1. World coordinate system
2. Camera Coordinate System
3. Image (planar) coordinate system
4. Sensor coordinate system



3

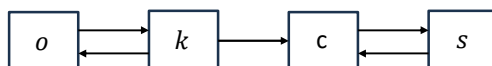
Camera Matrix

- To understand this sequence start with projection with what we already know
- The relationship between (x, y) and (X, Y, Z) has been linearized using homogeneous coordinates on both the left and right hand sides



$$\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$$

$$x = \frac{x'}{z'}, \quad y = \frac{y'}{z'}$$



Where are we in this picture?

$$x_{3 \times 1} = P_{3 \times 4} X_{cam} \text{ where } P = \text{diag}(f, f, 1) [I_{3 \times 3} | 0]$$

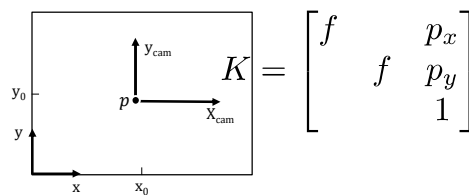
4

Camera Matrix

- We know

$$x_{3 \times 1} = P_{3 \times 4} X_{cam} \text{ where } P = \text{diag}(f, f, 1) [I_{3 \times 3} | 0]$$

- Principal point offset: the origin of the camera coordinate system is not aligned with the ideal physical camera
- Thus (accounting for translation) $x_{3 \times 1} = K_{3 \times 3} [I_{3 \times 3} | 0] X_{cam}$



Where are we in this picture?

5

CCD Camera Sensor Values

- $x_{3 \times 1} = K_{3 \times 3} [I_{3 \times 3} | 0] X_{cam}$
- We now move to pixel units

$$K = \begin{bmatrix} m_x & 0 & 0 \\ 0 & m_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & m_x p_x & 0 \\ 0 & \alpha_y & m_y p_y \\ 0 & 0 & 1 \end{bmatrix}$$

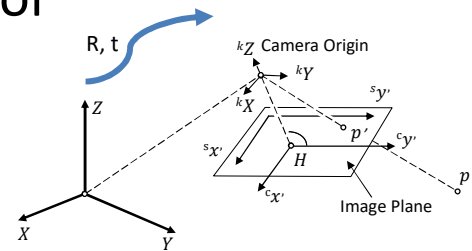


Where are we in this picture?

6

World to Sensor

- World coordinate system
- Camera Coordinate System
- $\hat{C}$ is the coordinate of the camera center in the world



$$X_{\text{cam}} = R(X - \hat{C})$$

$$X_{\text{cam}} = \begin{bmatrix} R & -R\hat{C} \\ 0 & 1 \end{bmatrix} X$$

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} = \begin{bmatrix} x \\ y \\ w \end{bmatrix} \approx \begin{bmatrix} 3 \times 1 \\ 3 \times 4 \\ 4 \times 1 \end{bmatrix} P \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$



Where are we in this picture?

7

Camera Matrix Components

- We move to pixel units but further model a non-ideal sensor (a skew between the two axis, i.e., the two axis are not necessarily at right angles)
- Redefined K is upper triangular
- **homogenous** $P = KR[I \mid -\hat{C}] = K[R \ t]$

$$X = \begin{bmatrix} 3 \times 1 \\ 3 \times 4 \\ 4 \times 1 \end{bmatrix} P \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$



Where are we in this picture?

$$K = \begin{bmatrix} \alpha_x & s & m_x p_x \\ & \alpha_y & m_y p_y \\ & & 1 \end{bmatrix}$$

3×3

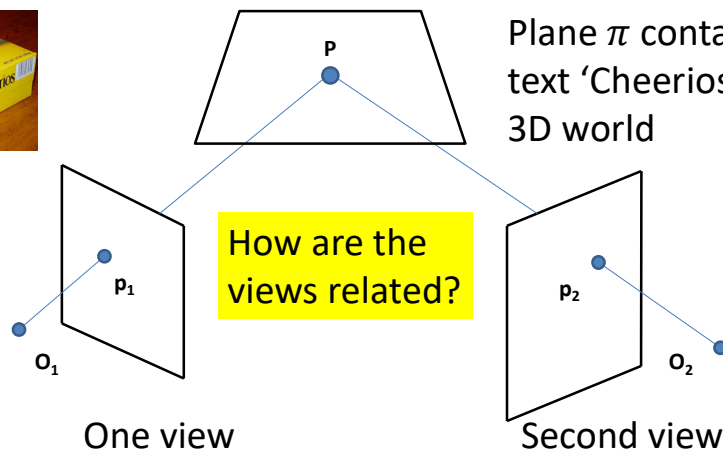
8

Agenda

- Dissecting the camera matrix

10

Views



If the second view is an "upright" view, then information will be preserved!

11

Planar Homography

- $x = PX$
- Recall: P can also be written as $P = K[R | t]$ where $t = -R \hat{C}$
- Arbitrarily place the world coordinate frame to correspond to $Z = 0$

Also insert the 3x3 invertible matrix to go to sensor values

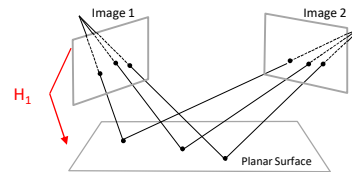
$$\lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$\lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix} \begin{bmatrix} X \\ Y \\ 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & t_x \\ r_{21} & r_{22} & t_y \\ r_{31} & r_{32} & t_z \end{bmatrix} \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} fr_{11} & fr_{12} & ft_x \\ fr_{21} & fr_{22} & ft_y \\ r_{31} & r_{32} & t_z \end{bmatrix} \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} \alpha_x & s & p_x \\ & \alpha_y & p_y \\ & & 1 \end{bmatrix}$$



12

Planar Homography

- $x = PX$
- Recall: P can also be written as $P = K[R | t]$ where $t = -R \hat{C}$
- Arbitrarily place the world coordinate frame to correspond to $Z = 0$

Also insert the 3x3 invertible matrix to go to sensor values

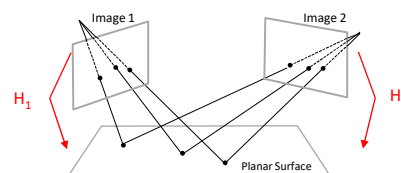
$$\lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$\lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix} \begin{bmatrix} X \\ Y \\ 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & t_x \\ r_{21} & r_{22} & t_y \\ r_{31} & r_{32} & t_z \end{bmatrix} \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} fr_{11} & fr_{12} & ft_x \\ fr_{21} & fr_{22} & ft_y \\ r_{31} & r_{32} & t_z \end{bmatrix} \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} \alpha_x & s & p_x \\ & \alpha_y & p_y \\ & & 1 \end{bmatrix}$$



13

Agenda

- Dissecting the camera matrix

$$\lambda_1 \begin{bmatrix} x_1' \\ y_1' \\ 1 \end{bmatrix} = H_1 \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix} = H_1^{-1} \lambda_1 \begin{bmatrix} x_1' \\ y_1' \\ 1 \end{bmatrix}$$

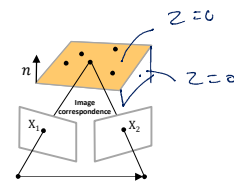
$$\lambda_2 \begin{bmatrix} x_2' \\ y_2' \\ 1 \end{bmatrix} = H_2 \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix} \Rightarrow \lambda_2 \begin{bmatrix} x_2' \\ y_2' \\ 1 \end{bmatrix} = H_2 H_1^{-1} \lambda_1 \begin{bmatrix} x_1' \\ y_1' \\ 1 \end{bmatrix}$$

$$\begin{matrix} \downarrow \\ \vec{X}_{\text{img}} \end{matrix} = \begin{matrix} \rightarrow \\ \tilde{H} \end{matrix} \begin{matrix} \vec{X}_{\text{orig}} \\ 3x3 \end{matrix}$$

15

Planar Homography

- Note: The distance between the two camera centers and the world point changes
- The world point however lies on the same plane



$$\lambda_1 \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix} = H_1 \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix} \Rightarrow \lambda \begin{bmatrix} x_2 \\ y_2 \\ 1 \end{bmatrix} = H_2 H_1^{-1} \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix}$$

Convert between projections of a 3D plane in 2 views
with a 3x3 matrix H

16

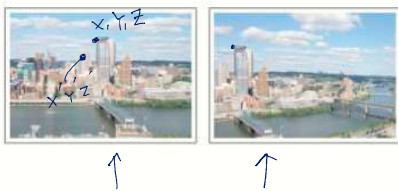
Agenda

- Dissecting the camera matrix

17

Bonus

Image Stitching



18

Pure Rotation Is a Homography

- Points don't line on the same plane
 - But the camera center does not move

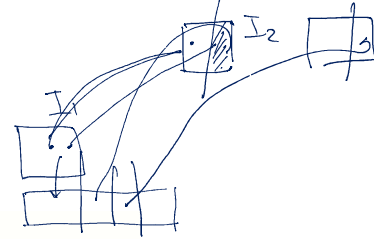


Image Stitching



19

Pure Rotation As Homography

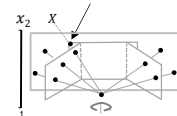
- What is new: we have a proof of the existence of a 3×3 homography

$$\lambda_1 \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

$$\lambda_2 \begin{bmatrix} x_2 \\ y_2 \\ 1 \end{bmatrix} = \begin{bmatrix} f_2 & 0 & 0 \\ 0 & f_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

$$\lambda \begin{bmatrix} x_2 \\ y_2 \\ 1 \end{bmatrix} = \underbrace{K_2 R_2 R_1^{-1} K_1^{-1}}_{H} \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix}$$

$H = 3K$



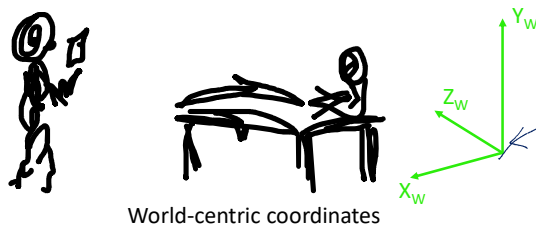
$$K = \begin{bmatrix} \alpha_x & s & p_x \\ & \alpha_y & p_y \\ & & 1 \end{bmatrix}$$

Sensor values (after inserting K)

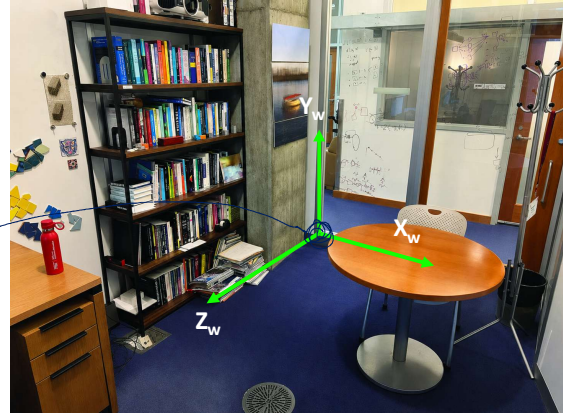
20

Agenda

- Dissecting the camera matrix



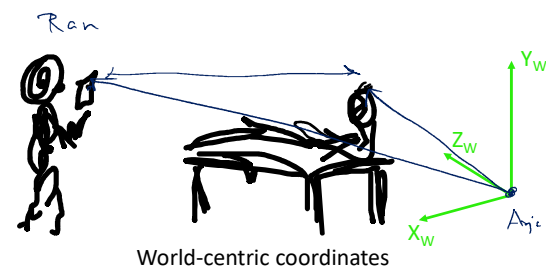
At what distance were you when you took the picture?



21

Dissecting a Camera Matrix

- If we have P
 - the null space of P corresponds to the origin of the camera in the world coordinate system.



$$P \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = 0$$

Handwritten notes include a vector $\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$ and a matrix P with dimensions 3×4 .

22

Dissecting a Camera Matrix

- The nullspace of P corresponds to the origin of the camera in the world coordinate system. Why?
 - Since P is 3×4 , it has a vector in the null space. Let the null space be generated by a vector C , i.e., $PC = 0$
 - Consider any point A in space and consider any point on the 3D parameterised line $X(t) = tA + (1 - t)C$. The image of any such a point under the transformation $x = PX$ is PA . This means that C must be the camera center.

23

Dissecting a Camera Matrix

- If we have P
 - the null space of P corresponds to the origin of the camera in the world coordinate system.



Assume you know the camera matrix
Where was this picture taken from?

24

Dissecting a Camera Matrix

- The first three column vectors of P can be interpreted as the vanishing point of the world axes
 - This follows because, for example, the y -axis has the direction $(0,1,0,0)^T$
- The world origin is imaged as the fourth column of P

$$P = \begin{bmatrix} a & p & d & c \\ b & q & e & r \\ c & r & f & s \end{bmatrix}$$

where $(p/r, q/r)$

